



Intern Final Presentation

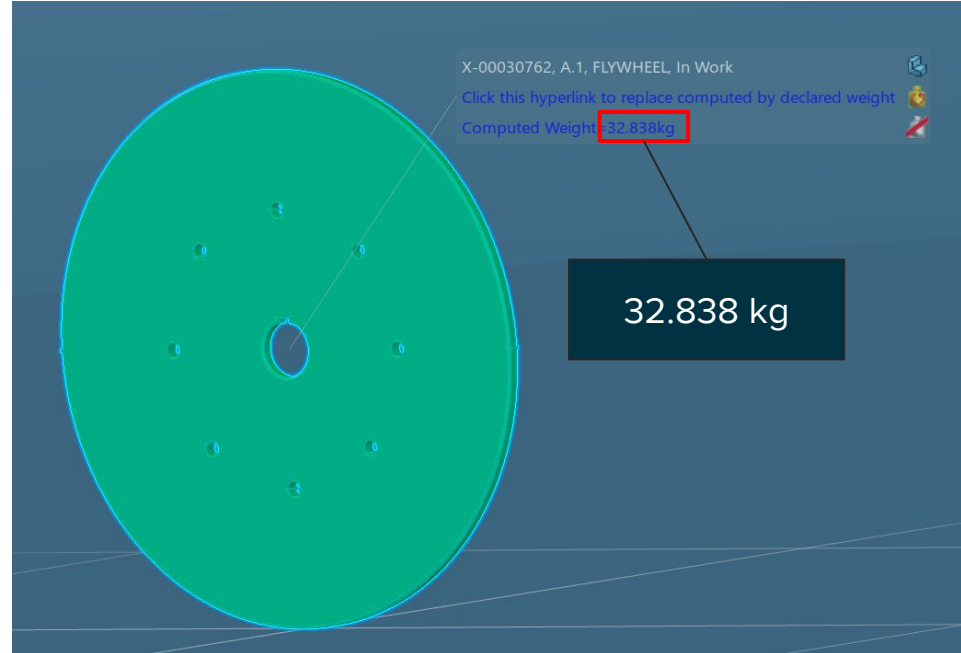
Sanjith Menon
8/28/26

Contents

- Park Lock Rig Inertial Load Dishrack
- Patch Lock Fastener Testing
- Electric Oil Pump NVH Testing
- Bolted Joint Review
- Clearcase Testing
- End-of-Line Oil Temperature Sensor Harness

Inertia Dishrack - Background and Problem

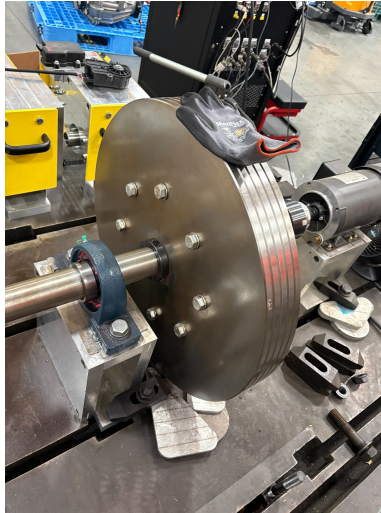
- Park Lock testing upcoming, requires additional inertial load capability
- Need storage for the disks and a method of loading the disks onto the shaft



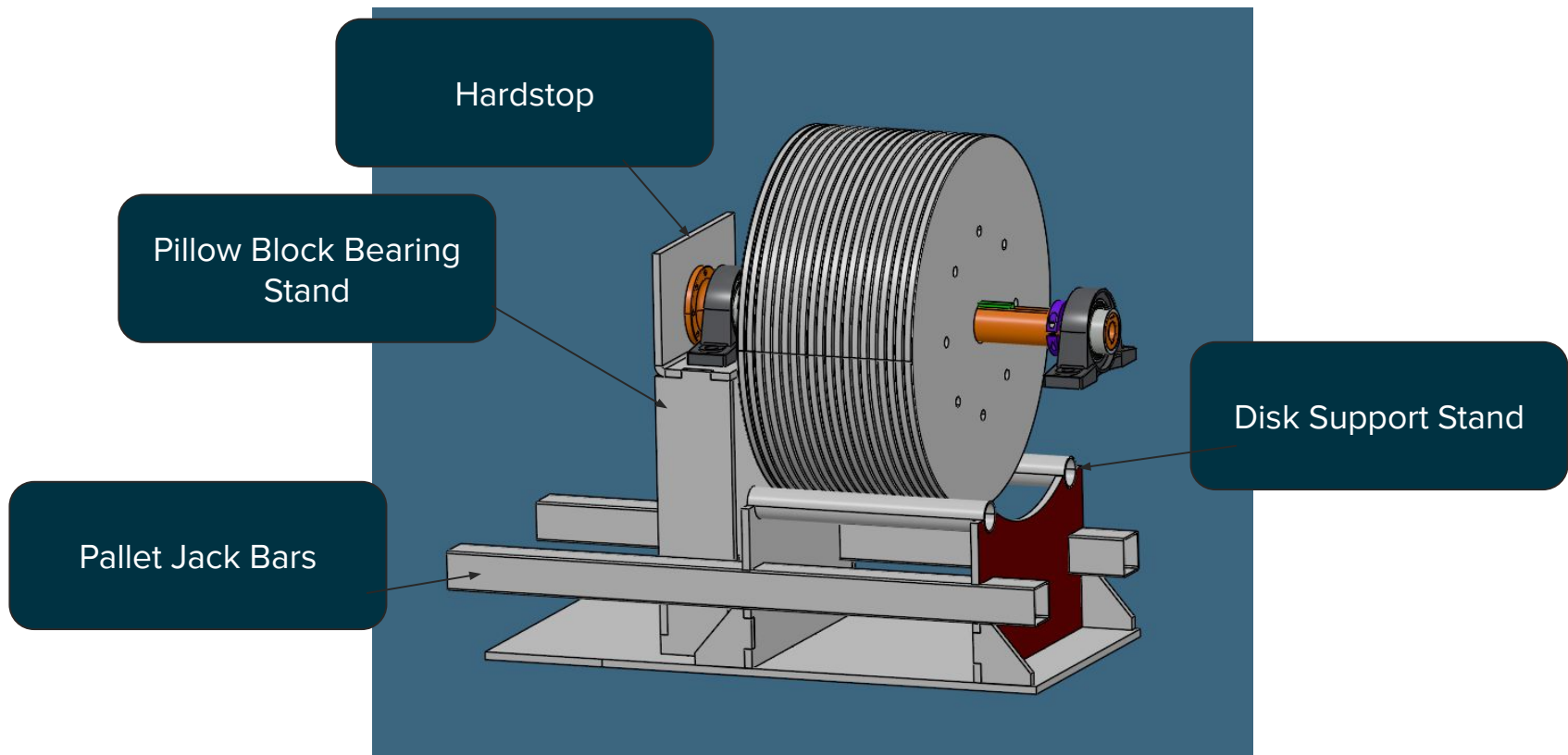
Total Load = 674 kg = 1490 lbs.

Inertia Dishrack - Solution

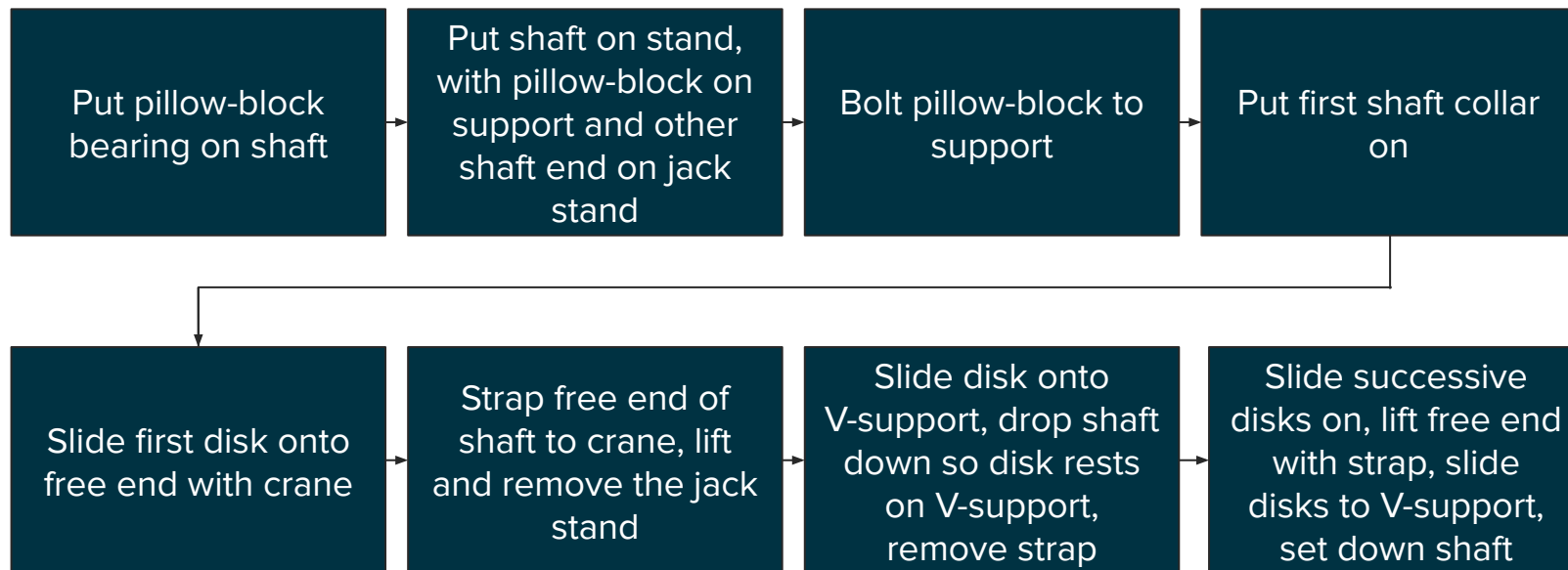
- Storage rack for disks x1
 - Must be bolted together so that can disassemble and store as flat plates if not needed
- Shaft loading tool x2
 - Must be able to store loaded and unloaded shafts
 - Must help make loading easier



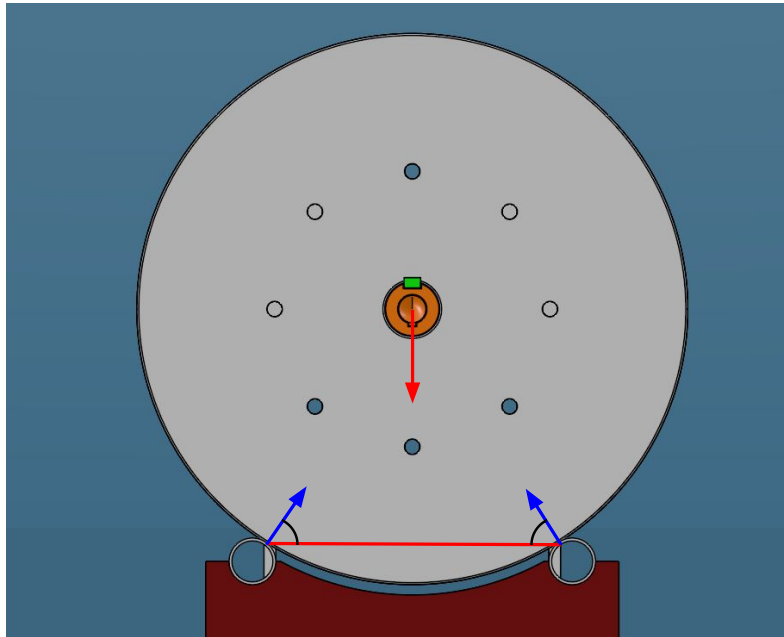
Inertia Dishrack - Shaft Loading Station



Inertia Dishrack - Shaft Loading Process



Inertia Dishrack - Shaft Loading Static Calcs



fully loaded force - 20 disks + shaft			
W	6605.2		
theta	1.005885608		
N	3910.084862		
Nx	2093.226426	Ny	3302.6
	for 1 disk	N1	195.5042431

After the load is determined, can
do the rest of the cool calcs

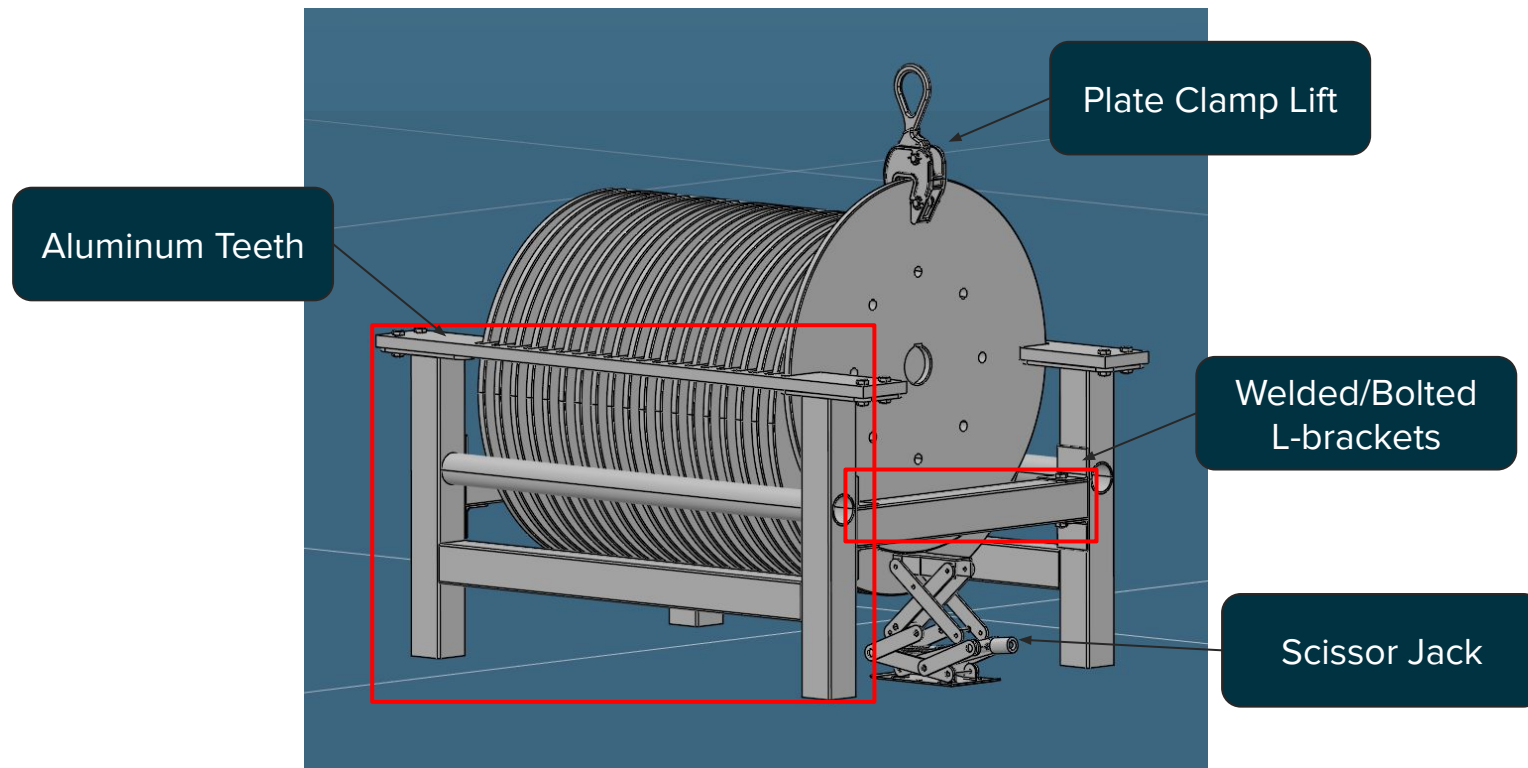
Inertia Dishrack - Shaft Loading Static Calcs

Failure Mode	Factor of Safety
Pallet Jack Tubes Bending	3.48
Contact Stress on Round Tube	23
Bearing Stand Horizontal Plate Bending	47
Bearing Stress in Plate-Tube Contact	58
Bending Stress on Round Tube	95
Support Plate Compression	263
Bearing Stand Vertical Plate Compression	905
Support Plate Buckling	8384
Bearing Stand Vertical Plate Buckling	9539



For fully loaded condition

Inertia Dishrack - Storage Rack



Inertia Dishrack - Storage Rack Static Calcs

Failure Mode	Factor of Safety
Fastener Bearing Stress in Hole	2.03
Teeth Body Tensile Yielding	8.3
Bolt Shear (Grade 8.8)	10
Bolt Tear-Out from Bar	10
Bolt Tear-Out from L-bracket	11
Tube Bending	12
Teeth Bending	15
Leg Compression	16
Crossbeam Tensile Yielding	25
Disk to Tube Contact Stress	29
Leg Buckling	17728

Inertia Dishrack - Key Learnings

- Hertzian Contact Stress, Bearing Stress calculations
- Tolerance Stackups
- Important to define the process and what is required to perform that process before designing
- Engineering Drawings



cylindrical contact stresses							
weight of disk	load	391 N	normal force	358.9533387			
angle of normal force		0.5759586532 radians	b	0.000010573439	0.000004610019	this is the depth at flip	
poissons		0.303 mild steel	Pmax	24858315.23			
disk diameter		0.6 m	sigmax	-15064139.03	at depth		
tube diameter		0.0508 m	sigmaz	-24858315.23			
E		2.00E+11 Pa	sigmay	-24858315.23			
L		0.869421 m	tmax	4897088.1	shear stress	Pa	
depth		0 m		4.8970881	MPa		
tensile strength		2.50E+08 Pa	fos	29.45628036			
shear strength	von mises criteri	144,250,000.00 Pa					
		144.25 MPa					

Patch Lock Testing - Overview

- Effort to convert from Loctite to Patch Lock
 - loctite uncontrolled amount applied
 - patch lock can be applied by supplier and arrives here ready to go
- Need to confirm no degrade
- Testing on diff bolts, park lock actuator, oil strainer, and resolver



Patch Lock Testing - Test Actions

Diff Bolt Testing - Atlas Copco Giant Tool

Install

Turn by 10° -
breakaway torque
test

Torque to yield

PLA, Oil Strainer, Resolver - Atlas Copco Wrenches

Install

Turn by 10° -
breakaway torque
test

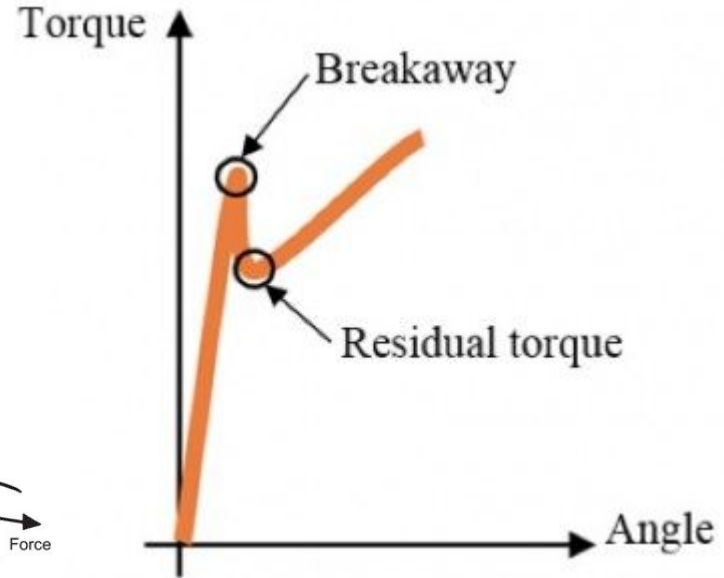
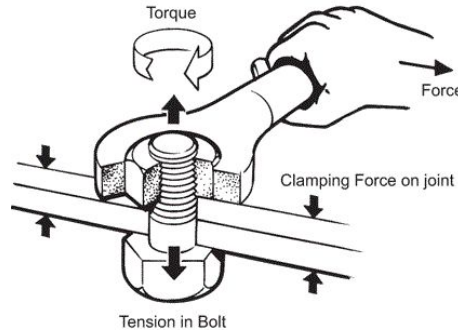
Uninstall and
Reinstall

Turn by 10° -
breakaway torque
test 2

Torque to yield

Patch Lock Testing - Interesting Concepts

- Breakaway Torque - torque required to break static friction of seated threads, indicator of likelihood of bolts backing out
- Yield Torque - torque applied for yielding to occur in bolt
 - torque goes into both overcoming friction and increasing preload on bolt
 - therefore yield torque changes with thread locker



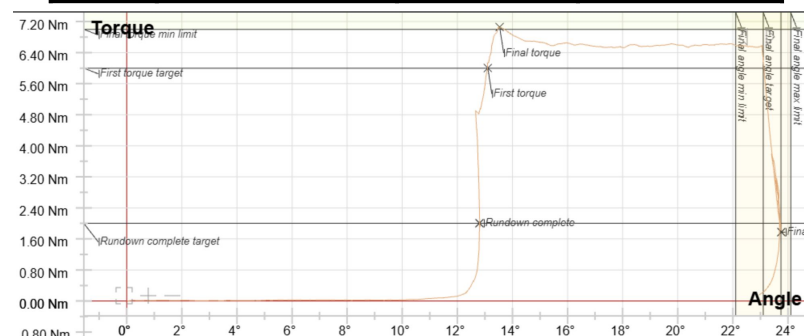
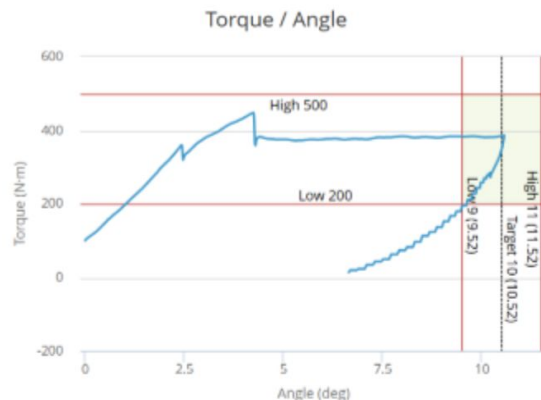
Patch Lock Testing - Results

DIFF BOLTS	Patch-Lock ed	Baseline	Verdict
Breakaway Torque Average (Nm)	436.995	448.87375	About the same
Average Peak Torque from Torque-to-Yield Test	855.9833333	758.26	Patch Lock

Locking	Sample #	Fastener Location	Install Torque 1	Breakaway 1	Residual 1	Install Torque 2	Breakaway 2	Residual 2	Torque @ failure
ND patch	Averages	Actuator	12.16	14.20	11.54	11.85222222	14.82333333	11.94	21.96
		Oil strainer	12.21	13.75	11.93	11.76	14.8	13.16	
		Resolver	6.02	5.53	5.3425	6.02	4.101666667	4.09	8.648333333

Locking	Fastener Location	Breakaway 1	Residual 1
Loctite 242	Actuator	15.95	12.10
	Oil strainer	13.64	12.83
	Resolver	5.374166667	5.186666667

PLA	Torque to Plastic Housing Failure
patch lock	14.498
Baseline (no thread locking mechanism)	13.74666667



Patch Lock Testing - Recommendations

Diff Bolt Recommendations:

Nominal Install Torque: **350 Nm** → **394 Nm**

Perform breakaway testing at new install torque

PLA, Oil Strainer, Resolver Recommendations:

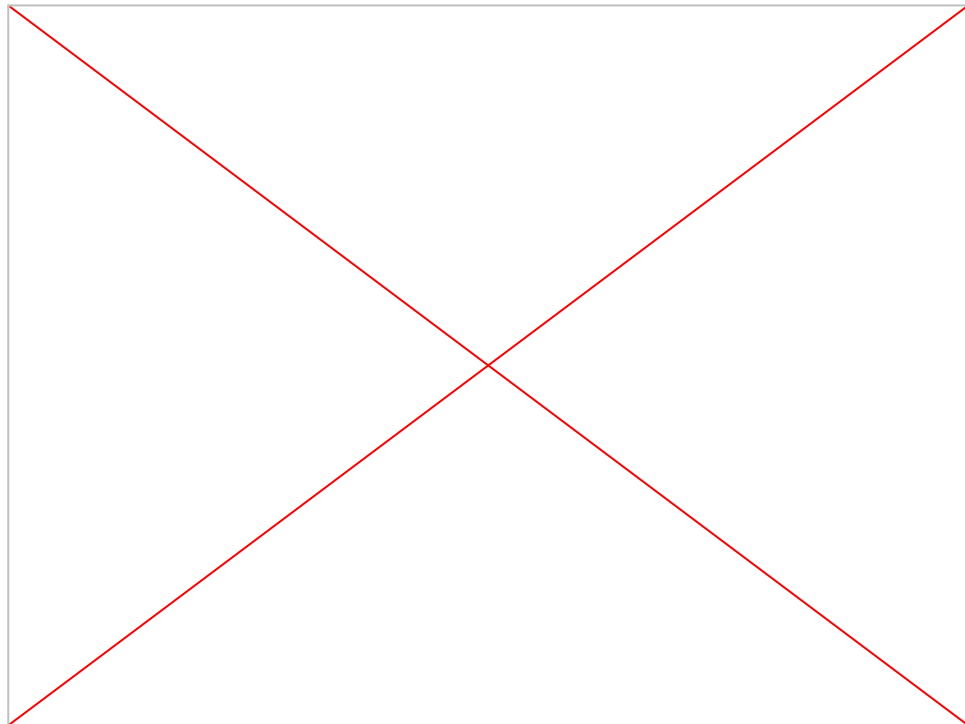
Loctite → **Patch Lock** **OK**

DO NOT reuse bolts - breakaway 2 is no good

Park Lock Actuator Install Torque: **No change**

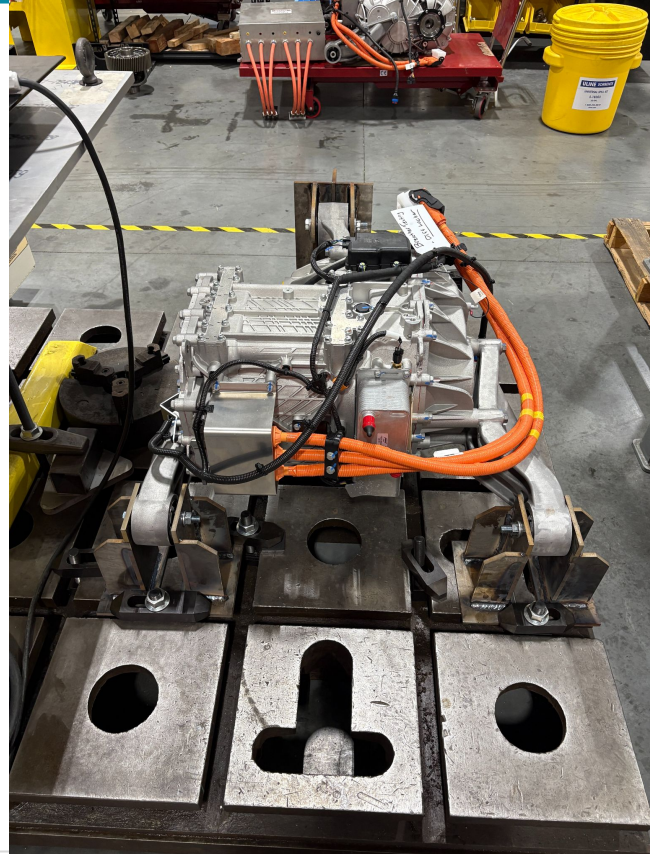
Resolver Install Torque: **6 Nm** → **6.4 Nm**

If possible **do not use manual** torque wrench for future



EOP NVH Testing - Background and Problem

- Variation in noise from oil pumps with cause unknown
- Testing required to collect information and identify root cause



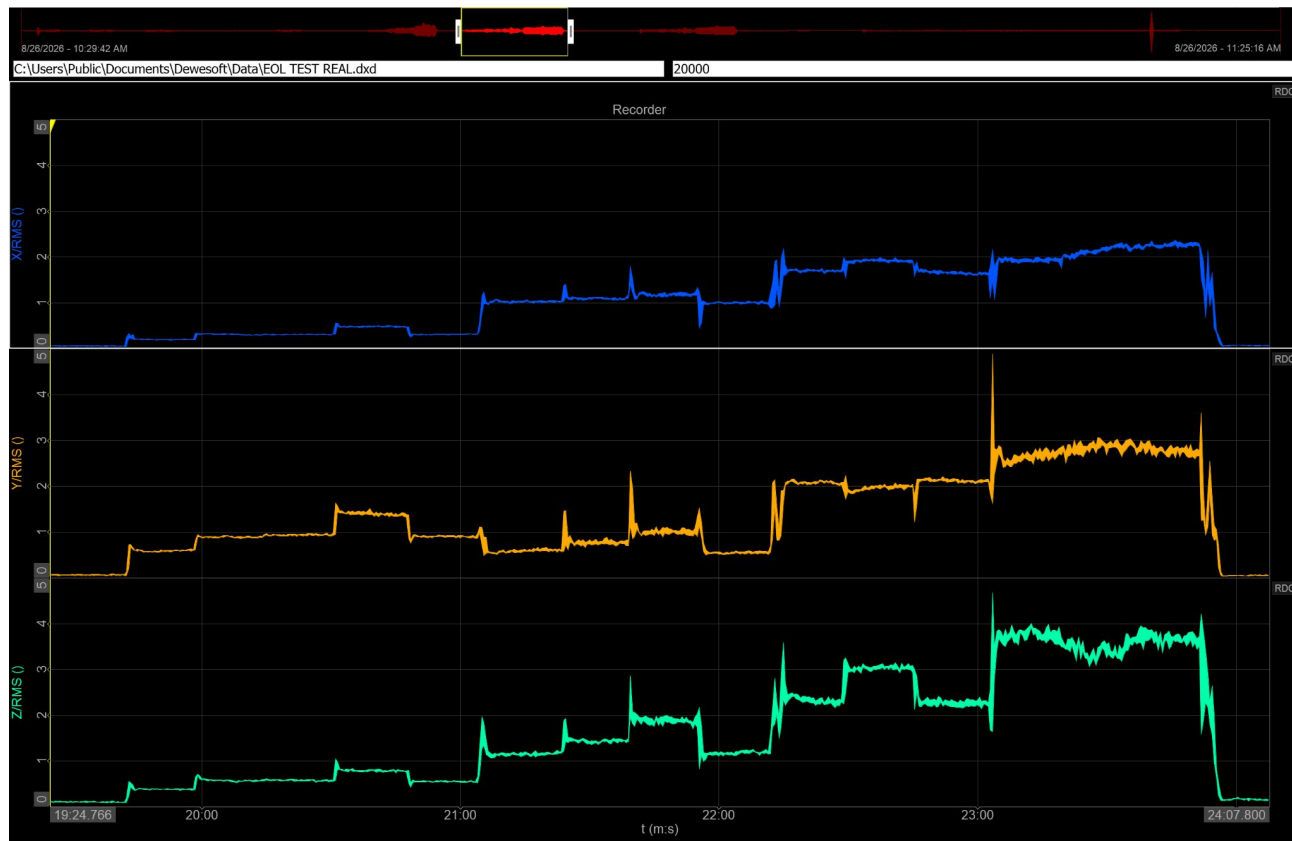
EOP NVH Testing - Plan and Setup

- Accelerometer positioning test
 - determine optimal location for accelerometer
 - factors include magnitude of vibrations, frequencies of vibrations in FFT, distinct modes, etc.
- Control group test
 - collect vibration data on a unit with a good pump for comparison
- Noisy pump EDU test
 - collect vibration data on a unit with a noisy pump, see which frequencies have higher peaks

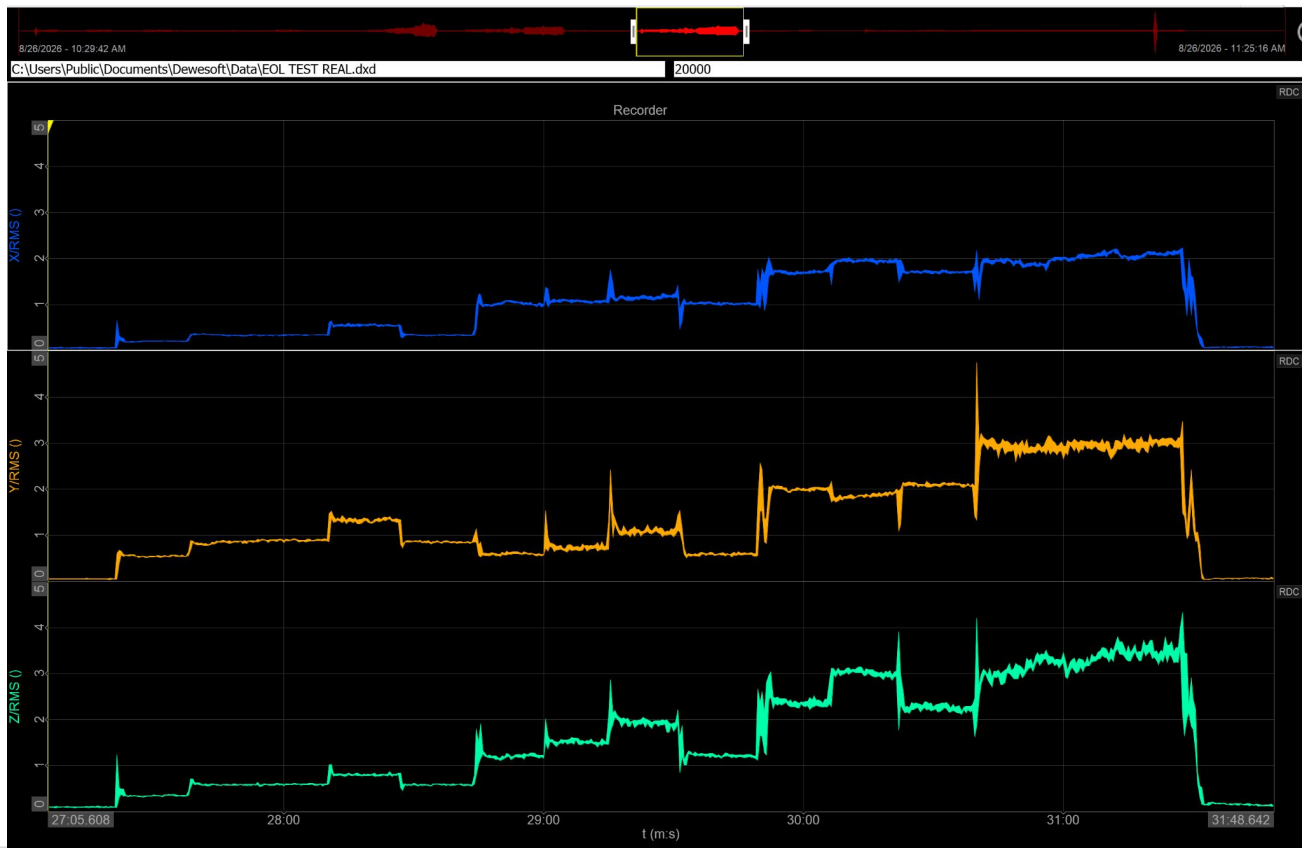
BUT Dewesoft DAQ has issues so tried validating against EOL accel data



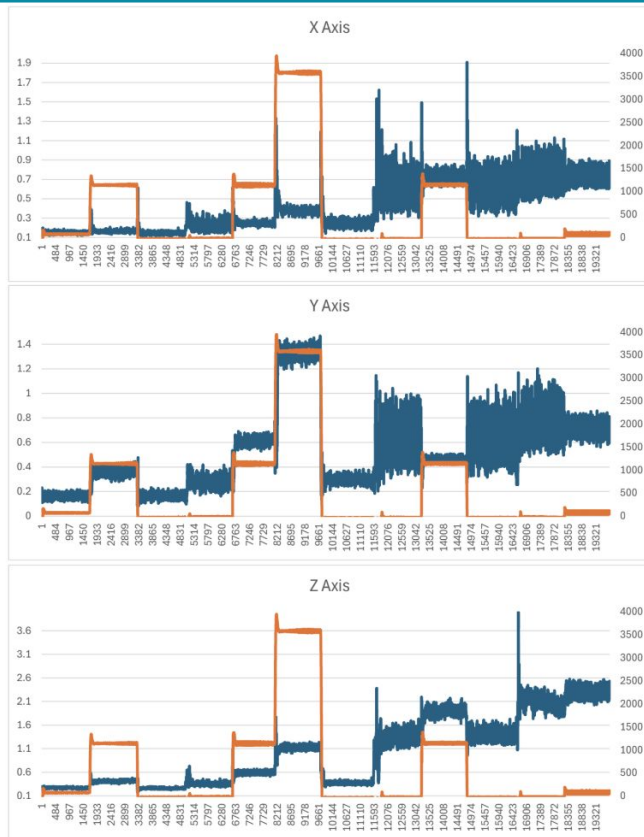
Accelerometer EOL Testing



Accelerometer EOL Testing



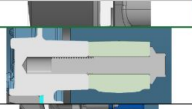
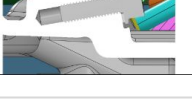
Accelerometer EOL Testing



Conclusion: safest option is to just send the DAQ to Dewesoft

Bolted Joint Review

- Reviewed all bolts in EDU to calculate clamp load, percentage of proof load, etc.
- Ties in with vehicle-wide audit of bolted joints
- Learned a lot about loading of fasteners, grip length and thread engagement rules of thumb
- Next Steps: torque testing on bolts - lots of no goods (friction assumptions), compression limiters

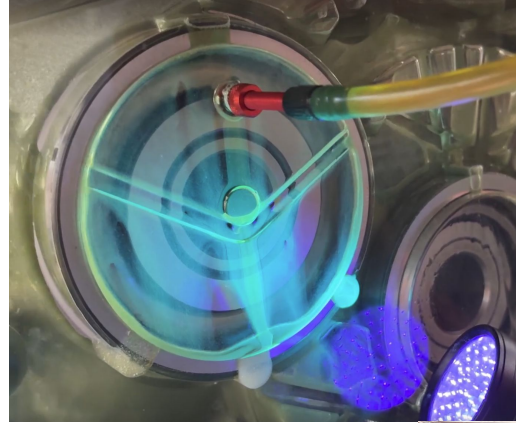
ASM No.	REV	JOINT	FASTENER PART NUMBER & REV	SUPPLIER / COATING	GRADE	SIZE	CROSS-SECTION	CLAMP LOAD		PROOFLOAD (N)			THREAD ENGAGEMENT (mm) (required)
								CLAMP LOAD (kN) achieved	CLAMP LOAD (kN) judgement	PROOFLOAD (N)	%	PROOFLOAD JUDGEMENT (65% - 75%)	
A-00182227	C	EA4 - FR LH BRACKET ASM TO GEARBOX ASM	H-00052902	COATING: ZnNi or ZnAl with topcoat generally recommended	10.9	M14-2.00		58.8	NG	95815	62.28%	NG	28.0
A-00182227	C	EA5 - FR RH BRACKET ASM TO ENDBELL ASM	H-00052902	COATING: ZnNi or ZnAl with topcoat generally recommended	10.9	M14-2.00		58.8	NG	95815	62.28%	NG	28.0
A-00182227	C	EA6 - RR BRACKET ASM TO MOTOR ASM	H-00052902	COATING: ZnNi or ZnAl with topcoat generally recommended	10.9	M14-2.00		58.8	NG	95815	62.74%	NG	28.0
A-00182227	C	EA7 - RESOLVER GROUND TERMINAL TO RESOLVER COVER ASM	H-00221018	COATING: ZnNi or ZnAl with topcoat generally recommended	8.8	M5-0.80		5.1	NG	8510	85.37%	NG	10.0

Clearcase Testing

- Addition of shielded bearing
- Removal of endcap from input shaft
- Visually inspected fluid flow at flat grades and extreme grades for different RPMs
- Conclusion/Next Steps:

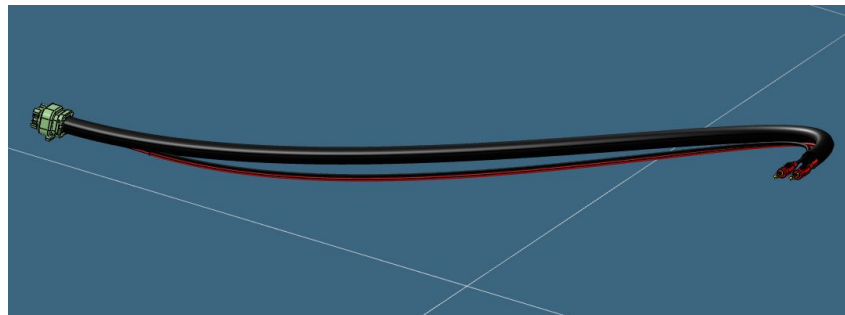
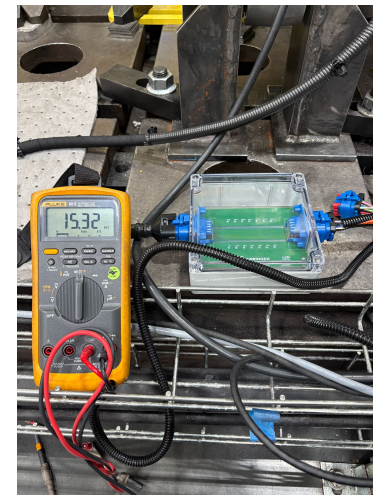
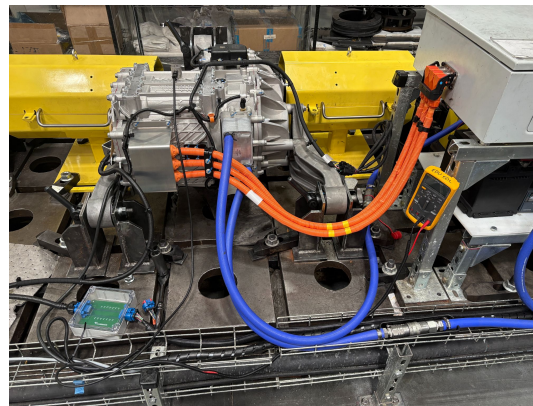
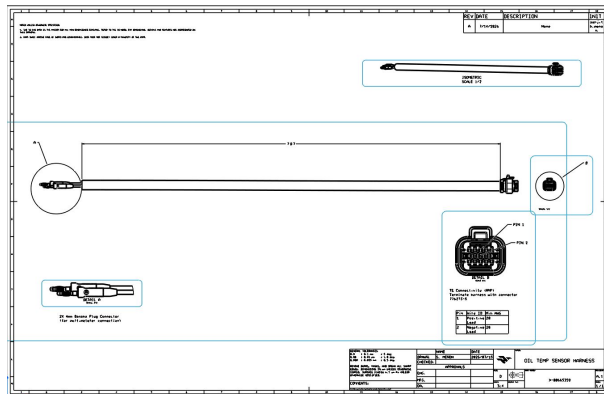
Shielded bearing is **effective** for reducing oil pooling.

Input spline sees **NO** oil, might have to do further studies into optimizing the endcap and testing those newer designs.



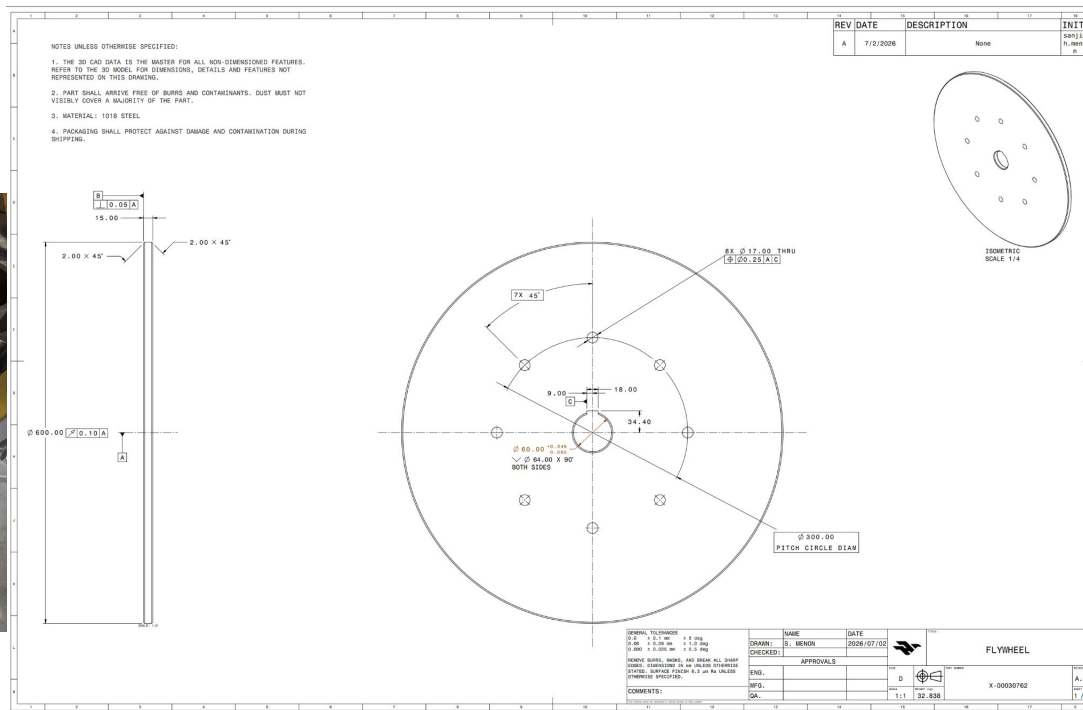
EOL Oil Temperature Sensor

- designed in CAD (released)
- made engineering drawing
- made 2 harnesses (1 spare)
- wrote work instruction changes
- it is now implemented in production process



Miscellaneous

- Inertia Flywheel DWG + sourcing
- Park Lock Rig build up components
- EDU stands





HARBINGER

THANK YOU